Cross-Channel Agreement Beats Consensus: Compositional Verification for Geometry Reasoning

Anonymous Author(s)
Affiliation
Address
email

Abstract

Multimodal geometry reasoning requires models to jointly perceive diagrams and perform symbolic derivations. Test-time verification methods such as majority voting assume voter errors are roughly independent, but in geometry this assumption breaks down: all candidates perceive the same diagram, so perceptual biases cascade into correlated errors that produce confident but wrong consensus. We observe that neuro-symbolic candidates contain compositional structure that can serve as a verification resource. We introduce SKETCH, a format decomposing each candidate into a neural pathway (natural-language claim) and a symbolic pathway (visual grounding $\rightarrow$ typed DSL $\rightarrow$ deterministic execution). These pathways are governed by different error processes—symbolic errors are correlated across candidates through shared perception, while neural errors are more dispersed. Our method, Compositional Consensus Verification (CCV), gates consensus on cross-channel agreement: because the two pathways fail for different reasons, accidental agreement is rare, making each concordant vote far more precise than a raw majority vote. Across four geometry benchmarks, CCV achieves 81.0% accuracy (+10.4 pp over majority voting) without external scorers or additional inference. Seven non-compositional baselines—including trained PRMs—cluster at 70–71%, while compositional cross-layer features jump to 79–81%, confirming that structural diversity is the core verification resource, and that a principled gating algorithm effectively unlocks its potential.

1 Introduction

The verification bottleneck in geometry reasoning. Geometry reasoning differs from purely textual mathematical tasks in a fundamental way: it requires joint visual perception and symbolic derivation. A model must read spatial relationships from a diagram—angles, parallel lines, congruences—and then chain these percepts through multi-step logical deductions. This conjunction makes geometry a task that current vision-language models find difficult Gao et al. [2023], Xia et al. [2024], Zhang et al. [2024b], and it also creates a distinctive failure mode for test-time verification. Figure 1 summarizes our proposed solution.

The standard approach for improving reasoning accuracy at test time is self-consistency Wang et al. [2022]: sample K candidates and select the most frequent answer via majority voting (MV). This rests on the Condorcet assumption that voter errors are approximately independent, so that the majority converges to truth as K grows. In geometry, this assumption breaks down in a specific and measurable way. All K candidates perceive the *same* diagram; if the model harbors a systematic perceptual bias—misidentifying a perpendicular as supplementary, or misreading an angle label—then a majority of candidates will share the same error. Votes become correlated because they reflect the same perceptual errors. Formally, when the pairwise error correlation ρ is substantial, the effective

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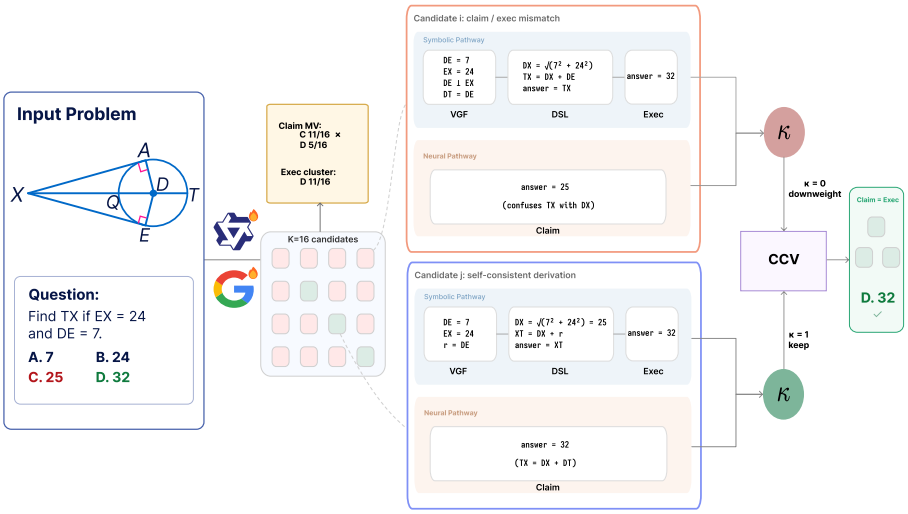


Figure 1: **Method overview.** A geometry problem is sampled into $K{=}16$ SKETCH candidates. Each candidate exposes a symbolic pathway (VGF $\rightarrow$ typed DSL $\rightarrow$ execution) and a neural pathway (claim). Cross-channel agreement κ filters reliable candidates; if the κ-pool is empty, CCV falls back to a high-confidence execution majority or claim majority vote.

sample size degrades as $N_{\text{eff}} = K/(1 + (K{-}1)\rho)$ Ladha [1992], Nitarach [2026]. At $\rho = 0.5$, sixteen candidates carry the information of barely two independent votes. In our experiments, wrong execution consensus clusters of size 8–16 account for 86% of all verification failures—strong majorities that are confidently wrong because they derive from the same misperceived geometric facts.

Compositional structure as a verification resource. Since the problem is within-channel error correlation, a natural response is to seek a structurally independent verification signal. We observe that neuro-symbolic reasoning candidates are not opaque votes. Unlike standard chain-of-thought outputs—a single text sequence mapping to a single answer—a neuro-symbolic candidate decomposes into layers with distinct computational character:

$$\text{Candidate} = \big(\underbrace{\text{VGF}}_{\text{perception}} , \underbrace{\text{DSL}}_{\text{logic}}, \underbrace{\text{exec}}_{\text{computation}} , \underbrace{\text{claim}}_{\text{intuition}} \big) \tag{1}$$

where VGF (Visual Grounding Facts) captures geometric relations extracted from the diagram, DSL is a typed program encoding the reasoning steps, exec is its deterministic execution result, and claim is the model's stated answer.

The claim and exec arise from *structurally different pathways*. The claim is produced by the neural pathway—the model's implicit reasoning, subject to the dispersed noise of autoregressive generation. The execution result is produced by the symbolic pathway—a deterministic computation on extracted premises, subject to the correlated noise of shared visual perception. These two pathways have different error correlation structures: neural errors are relatively independent across candidates, while symbolic errors are highly correlated because candidates see the same diagram.

Existing verification paradigms overlook this structural difference. *Consensus methods* (MV, self-consistency, CISC) count votes within a single output layer, making them vulnerable to within-layer error correlation. *Learned verifiers* (ORM, PRM, VisualPRM) train an external model to score candidates, which requires domain-specific training data and adds inference cost. Neither paradigm exploits the compositional structure that is already present in the candidates. We propose a third approach: **compositional verification**, which derives verification signal from cross-pathway agreement within the candidates themselves.

The cross-channel verification principle. We formalize the following principle. Consider two verification channels C_1 (claim) and C_2 (exec), each providing noisy evidence about the correct answer z^*. Let ρ_1, ρ_2 denote the pairwise error correlations within each channel.

Principle (Cross-Channel Verification). *When within-channel error correlations are high ($\rho_2 \gg 0$), a single instance of cross-channel agreement—one candidate where C_1 and C_2 independently converge to the same answer—provides stronger evidence for correctness than k instances of within-channel consensus, for any k bounded by the effective sample size $K/(1 + (K-1)\rho_2)$.*

Within-channel consensus can be inflated by shared errors (all candidates misread the same angle), but cross-channel agreement requires two structurally different noise processes to simultaneously produce the same answer. The probability of this occurring by chance is governed by the cross-channel error coupling, which we empirically measure at $P(\text{exec correct} \mid \text{claim correct}) \approx 0.225$—far lower than the within-channel consensus rates, confirming that the two pathways are only weakly coupled.

Contributions. This paper turns the internal structure of neuro-symbolic candidates into a test-time verification signal. We introduce the SKETCH format, which decomposes geometry reasoning into VGF, typed DSL, deterministic execution, and a neural claim while maintaining >99.9% format compliance with a 9B VLM. Building on this structure, we propose CCV, a training-free verifier that gates scoring and output routing through claim–execution agreement, improving over MV by $+10.41$ pp across four benchmarks with no additional inference. The same fixed protocol transfers to Gemma4-E4B, yielding $+7.84$ pp over Claim MV, and a failure taxonomy shows why: κ-gating recovers 88.5% of the dominant failure subtype where symbolic consensus fails but neural intuition holds.

2 The SKETCH Format

Visual grounding facts (VGF). For cross-candidate verification to work, visual observations must be comparable across candidates in a structured way. Free-form natural-language descriptions of a diagram are not reliably alignable; two candidates may describe the same geometric relationship with different wording. We therefore represent visual grounding as a list of atomic facts drawn from a closed vocabulary of relations:

$$f = (\texttt{relation}, \texttt{entities}), \quad \texttt{relation} \in \{\texttt{parallel}, \texttt{perpendicular}, \texttt{collinear}, \ldots\} \tag{2}$$

The closed vocabulary ensures parseability and enables direct cross-candidate comparison of perceived geometric structure.

Typed DSL. The DSL is a restricted Python subset that derives the answer from VGF-grounded premises using geometry helper functions and standard mathematical operations ($\sqrt{\cdot}$, trigonometric functions, π, etc.). Restricting the subset—no imports, no I/O, no loops—ensures termination and deterministic execution, which is essential for the symbolic pathway to produce a reliable verification signal. Every program terminates with an explicit `answer = ...` assignment, so the execution result can be extracted automatically without answer parsing.

Closed-loop structure. The key property of the SKETCH format is that each candidate carries an intrinsic verification signal: whether its neural claim agrees with its symbolic execution. This signal exists because the format closes the loop: VGF provides premises $\rightarrow$ DSL reasons from premises $\rightarrow$ exec produces a deterministic result $\rightarrow$ claim states the model's answer. When claim $\approx$ exec (compositional consistency, κ=1), the neural and symbolic pathways have independently converged, providing evidence that the answer is correct without any external reference.

Training. We fine-tune Qwen3.5-9B with LoRA (r=8, α=16) on 2,012 geometry examples derived from the Geometry3K Lu et al. [2021] training split. We use the geometry3k subset of NVIDIA Nemotron-Image-Training-v3 Blakeman et al. [2025], which provides 2,078 CoT annotations generated by Qwen3-VL-235B. Each CoT trace is reformatted into VGF+DSL structure by DeepSeek V4 Pro. The resulting model achieves format compliance >99.9%, VGF parse success >98.7%, and DSL execution success >94.5% on test data.

3 Compositional Consensus Verification

3.1 Verification Signal Taxonomy

Given K candidates $\{c_1, \ldots, c_K\}$, each decomposed as $c_i = (\text{VGF}_i, \text{DSL}_i, \text{exec}_i, \text{claim}_i)$, we define a hierarchy of verification signals ordered by the structural diversity they require:

Level 1: Within-layer consensus. Signals computed within a single output layer, aggregated across candidates:

$$S_{\text{claim}}(z) = |\{i : \text{claim}_i = z\}| \qquad \text{(surface consensus)} \tag{3}$$

$$S_{\text{exec}}(z) = |\{i : \text{exec}_i = z,\ \text{exec}_i \neq \perp\}| \qquad \text{(execution consensus)} \tag{4}$$

Traditional MV selects $\arg\max_z S_{\text{claim}}(z)$. Execution MV selects $\arg\max_z S_{\text{exec}}(z)$.

Level 2: Cross-layer agreement. Signals that compare outputs across structurally different layers. The core signal is compositional consistency:

$$\kappa_i = \mathbf{1}[\text{claim}_i \approx \text{exec}_i \wedge \text{exec}_i \neq \perp], \tag{5}$$

which checks whether a single candidate's neural and symbolic pathways agree. Additional cross-layer features (e.g., cross-candidate cross-layer agreement μ_i) appear in the learned ablation (Section 4.5) but are not part of the hand-designed algorithm.

Level 3: Agreement-gated consensus. Cross-layer agreement gates within-layer consensus:

$$G_\kappa(z) = \sum_i \mathbf{1}[\text{exec}_i = z] \cdot \kappa_i \tag{6}$$

Only execution votes from self-consistent candidates count. This is the signal that CCV's scoring function is built on.

3.2 The Cross-Channel Verification Advantage

For notational simplicity, this subsection writes claim–exec agreement as equality; in implementation, κ_i uses approximate numeric matching and excludes failed executions.

Theorem 1 (Cross-channel agreement as a precision filter)**.** *Let X_i be the claim answer and Y_i the executed answer for candidate i, and let z^* be the correct answer. Define $\kappa_i = \mathbf{1}[X_i = Y_i]$, $t_\kappa = P(X_i = Y_i = z^*)$, and $f_\kappa = P(X_i = Y_i \neq z^*)$. Then the accuracy of self-consistent candidates satisfies the identity*

$$p_\kappa \triangleq P(Y_i = z^* \mid \kappa_i = 1) = \frac{t_\kappa}{t_\kappa + f_\kappa}. \tag{7}$$

Under an idealized shared-perception model where the exec pathway depends on a latent visual state V, the claim pathway is conditionally independent of exec, and false agreement is dispersed across $M-1$ wrong answers, $t_\kappa = p_c p_e$ and $p_\kappa > p_e$ whenever $p_c > 1/M$.

The theorem isolates the central mechanism: κ-gating is primarily a *precision filter*. It does not require repeated votes in the same channel; it requires two structurally different pathways to converge to the same answer. When the answer space is large, as in numerical geometry, false self-consistency f_κ is rare, so a retained vote has much higher precision than a raw execution vote.

Theorem 2 (One-vs-rest SNR approximation)**.** *In a one-vs-rest binary correctness approximation, majority reliability depends on per-vote accuracy p, pool size n, and pairwise correctness correlation ρ through*

$$\text{SNR}(p, n, \rho) = \frac{p - \frac{1}{2}}{\sqrt{p(1-p)(1 + (n-1)\rho)/n}}. \tag{8}$$

The κ-filtered pool improves over raw exec voting whenever the precision gain from filtering outweighs the combined penalty of smaller pool size and higher correlation (full condition in Appendix J).

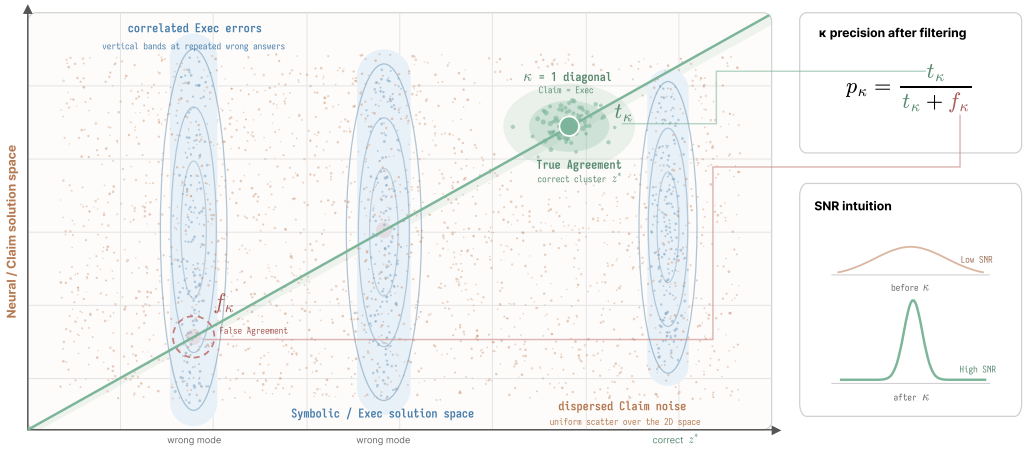


Figure 2: **Cross-channel precision filtering.** Left: κ-gating keeps candidates where claim and execution agree. True agreement occupies the claim-correct/exec-correct quadrant, while false agreement is dispersed across many wrong answers. Right: empirically, the κ-filtered pool is more correlated than raw execution votes, but its much higher precision yields a $4.1\times$ SNR improvement.

This yields a testable, counterintuitive prediction: κ-gating need not reduce correlation to improve majority reliability. Precision filtering can dominate the correlation penalty. This complements the negative findings of Nitarach [2026] on diverse prompting: when errors arise from shared perception, decorrelating samples is the wrong lever—filtering for cross-channel agreement targets precision directly.

Empirical support. We directly measure within-question correctness correlations on the full dataset (1,709 questions $\times$ 16 candidates). Raw exec votes have pairwise correlation $\rho_e = 0.526$; κ-filtered votes have higher correlation $\rho_\kappa = 0.646$. Yet per-candidate accuracy jumps from 61.7% to 87.1%, and the one-vs-rest SNR proxy improves by $4.1\times$ (0.32 to 1.33). Figure 2 visualizes this precision-filtering regime. Thus the retained pool is more correlated, but substantially more reliable—directly validating the counterintuitive prediction of Theorem 2: precision filtering dominates even when it increases correlation. This confirms that the dominant mechanism is precision filtering, not decorrelation: the $\kappa_i \cdot |\text{exec_cluster}_i|$ term in Algorithm 1 counts consensus only after it has survived cross-channel agreement.

3.3 κ-gated Dual-Pathway CCV

Algorithm 1 summarizes CCV. It uses claim-execution agreement as a precision gate, conceptually related to program-based consistency filtering Toh et al. [2024], but differs by combining agreement-gated consensus with output-pathway routing. The algorithm applies cross-channel agreement as a gate at two levels:

Candidate scoring (Lines 2–3). The term $\kappa_i \cdot |\text{exec_cluster}_i|$ realizes agreement-gated consensus: execution consensus is trusted only when accompanied by cross-layer agreement. The term $0.5 \cdot |\text{claim_cluster}|$ provides a baseline surface signal. Raw execution cluster size receives zero weight—our learned ablation in Section 4.5 confirms this design choice.

Output pathway routing (Lines 6–12). When self-consistent candidates exist ($\mathcal{C}_\kappa$ is non-empty), the algorithm outputs the neural claim. When cross-layer agreement collapses globally, it falls back to symbolic execution consensus, but only if the execution cluster is sufficiently large ($\geq \tau_{\text{exec}}$). This routing treats the *absence* of cross-channel agreement as diagnostic: it signals that the neural pathway is unreliable, making the deterministic symbolic pathway more trustworthy.

Complexity. $O(K)$ time, zero additional model calls, one tunable parameter (τ_{exec}).

Algorithm 1 κ-gated Dual-Pathway CCV

Require: Candidates $\{c_1, \ldots, c_K\}$, threshold τ_{exec}

for each candidate c_i **do**
  $\kappa_i \leftarrow \mathbf{1}[\text{claim}_i \approx \text{exec}_i \wedge \text{exec}_i \neq \perp]$
  $\text{score}_i \leftarrow 0.5 \cdot |\text{claim_cluster}_i| + \kappa_i \cdot (1 + |\text{exec_cluster}_i|)$
end for
$\mathcal{C}_\kappa \leftarrow \{c_i : \kappa_i = 1\}$ ▷ self-consistent candidate pool
if $\mathcal{C}_\kappa \neq \emptyset$ **then**
  return $\text{claim}(\arg\max_{c_i \in \mathcal{C}_\kappa} \text{score}_i)$ ▷ *Level 1: output neural claim*
else
  $z_{\text{exec}} \leftarrow \text{mode}(\{\text{exec}_i : \text{exec}_i \neq \perp\})$
  if $|\{i : \text{exec}_i = z_{\text{exec}}\}| \geq \tau_{\text{exec}}$ **then**
    return z_{exec} ▷ *Level 2: fall back to symbolic execution*
  else
    return $\text{mode}(\{\text{claim}_i\})$ ▷ *Level 3: fall back to claim majority*
  end if
end if

4 Experiments

4.1 Setup

Models and training. Our primary generator is Qwen3.5-9B, fine-tuned with LoRA (r=8, α=16) on 2,012 executor-validated Geometry3K training examples. The training data was constructed by reformatting Qwen3-VL CoT traces into the SKETCH format using DeepSeek V4 Pro. We also evaluate Gemma4-E4B to test cross-model generalization. Training was conducted using LLaMA-Factory with bf16 on 8 NVIDIA 4090 GPUs.

Table 1: Test benchmark suite. We evaluate on one in-distribution benchmark and three held-out/OOD settings.

Benchmark	Lang.	N	Role
Geometry3K test Lu et al. [2021]	EN	601	In-distribution
GeoQA test Chen et al. [2021]	ZH	754	Cross-lingual OOD
MathVista-GPS clean Lu et al. [2023]	EN	146	Cross-domain OOD
MathVista-GPS Lu et al. [2023]	EN	208	Reference

Evaluation protocol. Table 1 summarizes our evaluation suite. We use a frozen-rollout protocol with K=16 candidates, temperature 0.8, and top-p 0.95. All verification methods operate on the same rollout pool. The single hyperparameter τ_{exec}=9 was selected on a 300-question development split of Geometry3K and held fixed across all benchmarks and models.

4.2 Main Results

Figure 3 visualizes the same evidence ladder and K-scaling trend, while Table 2 reports the full benchmark breakdown. The four surface and confidence-based selectors—Claim MV, logprob-weighted MV, CISC, and URSA-RM weighted MV—all lie in a narrow 70.57–71.39% band, despite spanning self-confidence weighting and an external learned PRM. The learned ablation in Section 4.5 confirms this ceiling extends to logprob-based best-of-N and learned surface+execution feature selectors. Even URSA-RM, the strongest external scorer, achieves only $+0.76$ pp over Claim MV when used for weighted voting. Execution-only methods improve to 74–76%, but the large jump appears only when cross-layer agreement is used as a compositional signal: a PROVE-style κ filter reaches 79.52%, and CCV reaches 80.98%, a $+10.41$ pp improvement over Claim MV and $+4.80$ pp over EAV.

Paired per-question tests over the fixed 1,709-question evaluation set confirm that these main gains are statistically reliable: CCV improves over Claim MV by $+10.42$ pp (95% paired bootstrap CI

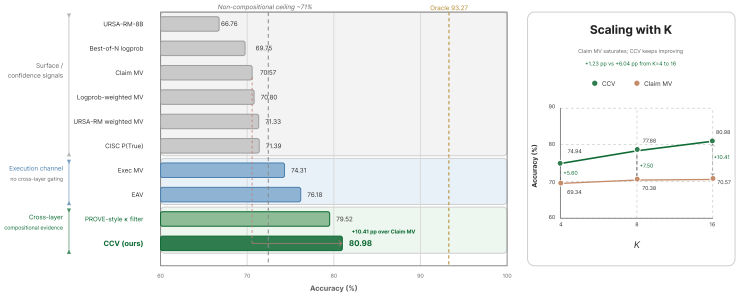


Figure 3: **Evidence ladder and K scaling.** Surface and confidence-based selectors cluster near 70–71%, execution-only methods improve moderately, and cross-layer compositional verification produces the main jump to 79–81%. As K increases, the CCV gain over Claim MV grows, indicating that compositional verification unlocks additional rollout diversity.

$[8.48, 12.40]$ pp; exact McNemar $p < 10^{-12}$) and over EAV by $+4.80$ pp (95% CI $[3.39, 6.26]$ pp; exact McNemar $p = 5.7\times10^{-11}$). The gain is also not concentrated on the development split: on the disjoint heldout/OOD split (excluding dev 300), CCV reaches 81.93%. Finally, a sweep over the only tunable parameter, τ_{exec}, shows that CCV is robust rather than finely tuned: across $\tau_{\text{exec}} = 5 \ldots 14$, dev accuracy varies by 1.00 pp and heldout/OOD accuracy by 1.06 pp (Appendix C). The κ matching threshold itself is also stable: varying the candidate-level relative tolerance around the default leaves CCV accuracy unchanged on the full s240 pool (Appendix H).

4.3 Cross-Model Generalization

To test whether CCV's gains depend on the specific base model, we repeat the full evaluation protocol with Gemma4-E4B, a 4B-parameter multimodal model with substantially weaker geometry reasoning capacity. The Gemma4 model achieves comparable SKETCH format compliance to the Qwen model, so this experiment is a confirmation of the verification mechanism rather than a new generator contribution.

Table E.1 reports the cross-model evidence ladder using the same τ_{exec}=9 for both models. First, the absolute accuracy gap is large: Claim MV drops from 70.57% on Qwen3.5-9B to 44.94% on Gemma4-E4B, consistent with the capacity difference. Second, the CCV gain pattern is preserved: CCV improves over Claim MV by $+10.41$ pp on Qwen and $+7.84$ pp on Gemma4. The Gemma4 result is especially diagnostic because its execution pathway alone is worse than the claim pathway (Exec MV 41.08% vs Claim MV 44.94%), making EAV ineffective (44.88%). Yet CCV still reaches 52.78%, showing that it exploits structural cross-channel agreement rather than the raw accuracy advantage of execution votes.

The mechanism predicted by Theorem 1 also replicates quantitatively: κ-filtering yields nearly identical precision gains across models ($+25.40$ pp vs $+26.38$ pp), and the counterintuitive correlation pattern repeats ($\rho_\kappa > \rho_e$ while SNR improves; Table D.1). This supports the interpretation that cross-channel precision filtering is model-agnostic. The CCV advantage also scales with K on both Qwen and Gemma4 (Appendix D, Appendix F).

4.4 K Scaling Analysis

Figure 3 shows that the CCV advantage scales with K: the gain over Claim MV grows from $+5.60$ pp at K=4 to $+10.41$ pp at K=16 (tabulated in Appendix D). Surface consensus saturates quickly (Claim MV gains only $+1.23$ pp from K=4 to 16), while compositional verification continues extracting signal from larger candidate pools.

As a negative control, VGF consensus—another within-layer signal—not only fails to help but *hurts* performance when added (from 80.05% to 79.93%). This reinforces the central finding: within-layer consensus—whether the layer is claim, execution, or VGF—is vulnerable to correlated perception errors. Only cross-layer gating provides robust verification signal.

Table 2: Evidence ladder across verification signal types (K=16). Surface/confidence methods, including learned PRM scores and self-judged confidence, cluster near 70–71%. Execution-only signals improve moderately. Cross-layer compositional verification produces the large jump to 79–81%.

Method	Type	G3K	GeoQA	GPS-c	GPS	All
Surface / confidence signals						
Claim MV	consensus	66.72	73.61	72.60	69.23	70.57
Logprob-weighted MV	weighted consensus	66.56	73.87	72.60	70.67	70.80
CISC $P(\text{True})$-weighted	confidence consensus	68.05	74.27	72.60	69.71	71.39
URSA-RM weighted MV	learned verifier	62.73	75.60	79.45	75.00	71.33
Execution channel without cross-layer gating						
Exec MV	exec consensus	72.71	75.60	76.71	72.60	74.31
EAV	exec-first priority	74.04	77.72	77.40	75.96	76.18
Cross-layer compositional signals						
PROVE-style κ filter	binary agreement	76.54	**82.10**	81.51	77.40	79.52
CCV (ours)	gated + routing	**78.54**	81.83	**86.30**	**81.25**	**80.98**
Oracle	upper bound	93.51	93.50	91.78	92.79	93.27

4.5 Learned Feature Ablation

To validate that the compositional features carry the transferable verification signal—not just that our specific CCV rule is effective—we train candidate-level logistic regression classifiers on Geometry3K dev (300 questions) and evaluate on disjoint OOD benchmarks.

Table D.3 reports the full ablation. The key result is that compositional features are the only transferable signal: logprob best-of-N, surface features, and surface+execution features remain near Claim MV (71.11–71.86% OOD), while adding cross-layer features jumps to 79.85% OOD and C-only features reach 81.02%. The learned model rediscovers the gating principle: raw exec_cluster_size has a negative coefficient (-2.40), while $\kappa \times$ exec_cluster is strongly positive ($+1.96$). A simple rule is therefore enough: PROVE-style κ filtering reaches 81.02% OOD, and dual-output CCV reaches 81.93% OOD.

4.6 Error Taxonomy and Recovery Analysis

We analyze 355 cases where EAV fails but an oracle-correct candidate exists (full flow in Figure D.1). Wrong execution consensus accounts for 305/355 = 86% of these failures. The largest recoverable subtype is Exec✗Claim✓: the execution majority is wrong, but the claim majority is correct. CCV recovers 123/139 = **88.5%** of this subtype by routing through candidates that survive claim–execution agreement. In contrast, Exec✗Claim✗ cases, where both pathways support the same incorrect answer, remain mostly outside the reach of selection-only verification (8.3% recovered). Among self-consistent candidates, reliability rises from 19% at execution cluster size 1 to 93% at size 8+, confirming that κ must be combined with gated consensus rather than used as a standalone binary filter.

5 Related Work

Geometry reasoning systems mainly improve the generator side of the problem. Specialized VLMs such as G-LLaVA Gao et al. [2023], GeoX Xia et al. [2024], and MAVIS Zhang et al. [2024b] strengthen multimodal geometry solving on benchmarks such as Geometry3K Lu et al. [2021] and GeoQA Chen et al. [2021]. Neuro-symbolic solvers such as AlphaGeometry Trinh et al. [2024], Chervonyi et al. [2025] and FormalGeo Zhang et al. [2023] use symbolic structure to derive proofs or answers. Our work is complementary: it assumes a candidate pool already exists and studies how to verify candidates when shared diagram perception makes their errors correlated.

Test-time verification is usually based on either consensus or learned scoring. Self-consistency Wang et al. [2022], universal self-consistency Chen et al. [2023], and confidence-weighted CISC Taubenfeld et al. [2025] aggregate opaque answer votes, which works best when sampled errors are suffi-

ciently independent. Learned verifiers, including ORMs/PRMs Cobbe et al. [2021], Uesato et al. [2022], Lightman et al. [2023], Wang et al. [2024], GenRM Zhang et al. [2024a], URSA-RM Luo et al. [2025], and VisualPRM Wang et al. [2025], train external scorers and add verifier-specific data or inference. CCV differs from both paradigms by deriving verification signal from structure already present inside each candidate.

The closest line of work uses programs or error-correlation analysis to diagnose voting failures. PROVE Toh et al. [2024] filters candidates whose translated Python output matches the natural-language answer, a binary consistency signal analogous to κ. Our framework generalizes this idea to geometry-specific neuro-symbolic outputs, where VGF, typed DSL, deterministic execution, and neural claims form a hierarchy of cross-layer signals. Nitarach [2026] show that correlated errors limit majority voting; CCV targets this regime directly by gating consensus through cross-channel agreement rather than trying to decorrelate samples.

6 Conclusion

We have shown that neuro-symbolic reasoning candidates carry an underutilized verification resource: their compositional structure. By formalizing cross-channel agreement as a verification principle, we derive a training-free algorithm that substantially outperforms both standard consensus methods and learned verifiers. The key insight—that agreement between structurally diverse pathways provides stronger evidence than repetition within any single pathway—is validated by a full evidence ladder: non-compositional confidence, consensus, and learned-score methods saturate around 70–71%, while κ-based compositional methods reach 79–81%. More broadly, output format is a verification design choice: domains that can expose perception, logic, computation, and conclusion as cross-validatable pathways may benefit from the same principle. Our current validation focuses on open-weight geometry models; extending CCV to proprietary frontier models and adding independent premise verification are natural next steps.

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A Training Data Construction

Table A.1 summarizes the full construction pipeline for the 2,012-example SFT set. We start from the official Geometry3K Lu et al. [2021] training split, use the geometry3k subset of Nemotron-Image-Training-v3 Blakeman et al. [2025] as the Qwen3-VL-235B-A22B-Thinking CoT source, reformat each matched trace into the SKETCH VGF+DSL+answer format with DeepSeek V4 Pro, and retain only examples whose DSL executes and matches the official answer. The reformatting stage is text-only: the expensive visual reasoning is supplied by the Qwen3-VL CoT trace, while the official gold answer is used as the final target for machine-checkable reformatting.

Table A.1: Training data construction and filtering. Percentages are relative to the 2,078 matched CoT rows unless otherwise noted.

Stage	Count	Rate
Official Geometry3K train rows	2,101	–
Matched Qwen3-VL CoT traces	2,078	98.91% of official
Qwen3-VL boxed-answer correct	2,065	99.37%
DeepSeek V4 Pro reformat attempts	2,078	100.00%
Strict validator pass	1,963	94.47%
Recovered by tolerant VGF normalization	49	2.36%
Final executor-validated SFT rows	2,012	96.82%
Filtered: execution-answer mismatch	61	2.94%
Filtered: DSL execution/safety error	5	0.24%

The final 66 filtered examples consist of 61 execution-answer mismatches and five DSL execution/safety failures: one undefined `UB`, one undefined `z`, one undefined `m`, and two blocked-token errors. The tolerant VGF pass clears unsupported or over-arity visual facts only when the DSL and answer remain executor-valid; a final post-processing pass keeps 2,012 rows, with 41 DSL snippets changed by the sanitizer and 6 VGF lists cleared.

B Data Construction Prompts

Stage 1: stored source prompt for frontier CoT traces. We did not rerun Qwen3-VL-235B-A22B-Thinking for this paper. We use an existing CoT source whose stored user messages follow the problem image plus the following prompt pattern:

```
Solve this geometry problem. Study the diagram and select
the correct answer.

{problem_text}
{choices}

Do not explain your answer but write only the answer label
(A, B, ...) in this format:
Answer: \boxed{...}.
```

Stage 2: VGF+DSL reformatting. DeepSeek V4 Pro receives no image input. It reformats the official problem, choices, official answer, and Qwen3-VL CoT into the target SKETCH format using the following template:

```
Convert the provided Qwen3-VL-235B natural-language geometry
chain-of-thought into the target VGF+DSL format.
Do not solve from scratch. Use the teacher CoT as the visual
and reasoning evidence, and use the official gold answer as
the final target.
Return exactly three sections in this order and no extra sections:
[VGF]
[DSL]
```

```
[Final Answer]

[VGF] rules:
Output a raw JSON list only. Do not use markdown fences.
Each item must be {"relation": <relation>,
"entities": [<tokens>]} with exact relation arity.
Allowed relations: {allowed_vgf_relations}.
Relation arities: {relation_arities}.
For unary shape relations, put the whole object in one entity
token, e.g.
{"relation":"parallelogram","entities":["MNPQ"]}.
Use at most 4 facts to stay safely under the validator limit
of 5 facts.
Use [] if no reliable visual fact is needed.

[DSL] rules:
Output executable Python assignment statements only.
Do not use markdown fences.
Do not include comments or prose.
Never write import statements, including 'from math import ...'.
Math functions are already available by name.
Every variable must be assigned before it is used.
Do not write placeholder equations with unknown symbols.
The final assignment must define a variable named answer.
The value of answer must evaluate to the official gold option
value; use the teacher CoT equations to derive that value.
Allowed math names: sqrt, pi, sin, cos, tan, radians, degrees,
asin, acos, atan, abs, min, max, round.
Do not read files, import modules, call external tools, or use
input/output functions.
[Final Answer] must be exactly the official gold option letter:
{gold}.
In [DSL], set answer to the numeric or symbolic value of
option {gold}:
{gold_value}.

Official problem:
{problem_text}
{choices}

Official gold answer: {gold}
Official gold option value: {gold_value}
Teacher boxed answer: {qwen_boxed_answer}

Teacher NL CoT:
{qwen_cot}
```

C τ_{exec} Sensitivity

The only tunable hyperparameter in CCV is the execution-majority threshold τ_{exec}, used only when the κ-pool is empty. We sweep $\tau_{\text{exec}} = 1 \ldots 16$ on the main Qwen3.5-9B s240 K=16 rollout pool, using the same split protocol as the main paper: Geometry3K qid<300 for dev, and Geometry3K qid≥ 300 plus GeoQA, MathVista-GPS clean, and MathVista-GPS for heldout/OOD evaluation. Figure C.1 plots the full sweep, and Table C.1 reports selected points. We use the resulting τ_{exec}=9 for both Qwen3.5-9B and Gemma4-E4B.

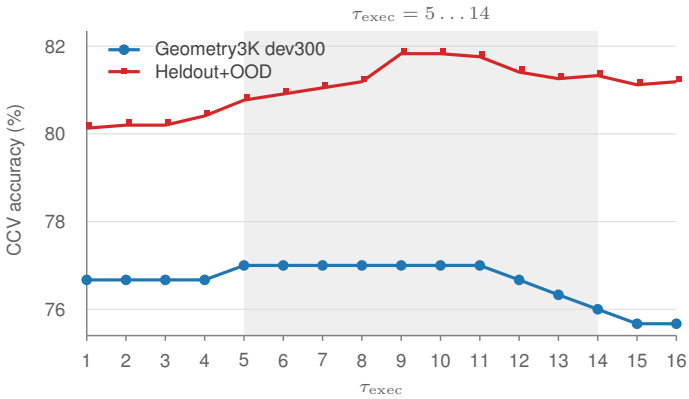


Figure C.1: **Sensitivity to τ_{exec}.** CCV is stable over a broad range of execution thresholds. In the plateau $\tau_{\text{exec}} = 5 \ldots 14$, dev accuracy varies from 76.00% to 77.00%, and heldout/OOD accuracy varies from 80.77% to 81.83%.

Table C.1: Selected points from the τ_{exec} sweep (full curve in the figure above). The dev-selected threshold is $\tau_{\text{exec}} = 9$.

τ_{exec}	Dev@300	Heldout/OOD	All
1 (no threshold)	76.67	80.13	79.52
9 (selected)	**77.00**	**81.83**	**80.98**
16 (maximum)	75.67	81.19	80.22

D Additional Diagnostics

This appendix contains the diagnostic tables and flow visualization referenced by the main experimental analysis. Table D.1 reports the precision-filtering statistics behind the cross-model claim, Table D.2 gives the exact Qwen K-scaling values summarized in Figure 3, Table D.3 gives the learned feature ablation, and Figure D.1 visualizes the error taxonomy and routing behavior.

Table D.1: Cross-model diagnostic for the precision-filtering mechanism. The κ-filtered pool is more correlated than the raw execution pool, but its much higher per-candidate precision improves the effective verification signal.

Model	p_e (%)	p_κ (%)	Δp (pp)	ρ_e	ρ_κ	mean K_κ	SNR raw$\rightarrow\kappa$
Qwen3.5-9B	61.70	87.10	+25.40	0.526	0.646	9.09	0.32$\rightarrow$1.33
Gemma4-E4B	33.43	59.81	+26.38	0.504	0.630	5.77	$-$0.48$\rightarrow$0.24

E Cross-Model Validation

Table E.1 evaluates the same verification protocol on a lower-capacity Gemma4-E4B generator, using the same τ_{exec}=9 as Qwen3.5-9B. The Gemma4 pool uses the same frozen K=16 four-benchmark setting as the main Qwen3.5-9B experiments: Geometry3K test (601), GeoQA test (754), MathVista-GPS clean (146), and MathVista-GPS (208), for N=1709 questions and 27,344 candidates. The evaluation uses the benchmark files in `clean/assets`; it does not use the deprecated 400-question checkpoint-selection set.

The notable difference is that Gemma4 reverses the raw pathway ordering: Exec MV is below Claim MV (41.08% vs 44.94%), whereas Qwen's execution pathway is stronger than its claim pathway. Consequently, EAV, an execution-first priority rule, is ineffective on Gemma4 (44.88% vs 44.94% Claim MV). This pattern is expected for a smaller model with weaker visual grounding: within-layer execution consensus is more easily dominated by correlated perception errors. The result therefore strengthens rather than weakens the verification story. CCV does not trust the raw execution vote; it

Table D.2: Performance across K on the four-benchmark Qwen3.5-9B pool.

K	Claim MV	Exec MV	CCV	Oracle	CCV − Claim MV
4	69.34	68.98	74.94	84.74	+5.60
8	70.38	72.43	77.88	89.89	+7.50
16	70.57	74.31	80.98	93.27	+10.41

Table D.3: Compositional feature ablation. "S" = standard features (logprob, claim cluster), "E" = execution features, "C" = compositional cross-layer features (κ, $\kappa \times$ exec_cluster, μ, $\kappa \times \mu$, sc_rate). The last column reports OOD gain over Claim MV.

Method / Features	Dev@300	OOD	All	Δ OOD vs Claim MV
Claim MV (reference)	66.67	71.77	70.57	0.00
Best-of-N (logprob)	66.00	71.11	69.75	−0.67
LogReg S-only	67.00	71.86	70.74	+0.08
LogReg S+E	66.33	71.69	70.63	−0.08
LogReg S+E+C	74.67	79.85	78.47	+8.08
LogReg C-only	75.33	81.02	79.64	+9.24
PROVE-style κ filter (rule)	75.00	81.02	79.52	+9.24
Dual-output CCV (ours, rule)	77.00	81.93	80.98	+10.16

uses claim–execution agreement as a precision filter. Even when the execution pathway is worse in aggregate, candidates where claim and execution agree remain informative, yielding $+7.84$ pp over Claim MV and $+7.90$ pp over EAV on Gemma4.

F Gemma4 K Scaling

Table F.1 repeats the K-scaling analysis on the frozen Gemma4-E4B K=16 pool by subsampling candidates; K<16 rows are averaged over four random subsamples. We keep τ_{exec}=9 fixed to match the cross-model comparison in the main text.

The Gemma4 gain increases with candidate budget, from $+4.87$ pp at K=4 to $+7.84$ pp at K=16, matching the qualitative scaling trend observed for Qwen.

G Gemma4 Implementation Details

The Gemma4-E4B experiment uses the frozen checkpoint-720 LoRA adapter and the frozen rollout pool described in Appendix E; no additional model calls are made for the verification comparisons. The adapter uses LoRA rank 8 and $\alpha = 16$. On a 400-example format audit, the model reaches 96.0% complete-section rate, 99.0% VGF parse success, 96.0% DSL syntax success, and 94.25% DSL execution success, comparable to the format reliability of the Qwen model.

H κ Matching Threshold Sensitivity

The relation $X_i \approx Y_i$ is implemented by normalizing the claimed final answer and executed DSL answer, then applying numeric tolerance. Floating-point answers, fractions, square roots, and π expressions are parsed to real values; degree symbols are treated as unit suffixes and removed before numeric comparison. For multiple-choice questions, numeric answers are first mapped to option letters using the fixed choice tolerance 0.02. Table H.1 reports an ablation that keeps the numeric absolute tolerance fixed at 10^{-3} and sweeps only the candidate-level relative tolerance used for κ.

All threshold settings yield identical results because geometry answers are effectively discrete: integers, simple fractions, radicals, angles, or multiple-choice letters. In this domain, claim–execution agreement is almost always binary: pairs either match exactly, or differ by margins much larger than even the loosest tolerance. The matching tolerance is therefore not a meaningful hyperparameter for these benchmarks, and the measured CCV gain is not an artifact of finely tuned numeric equality.

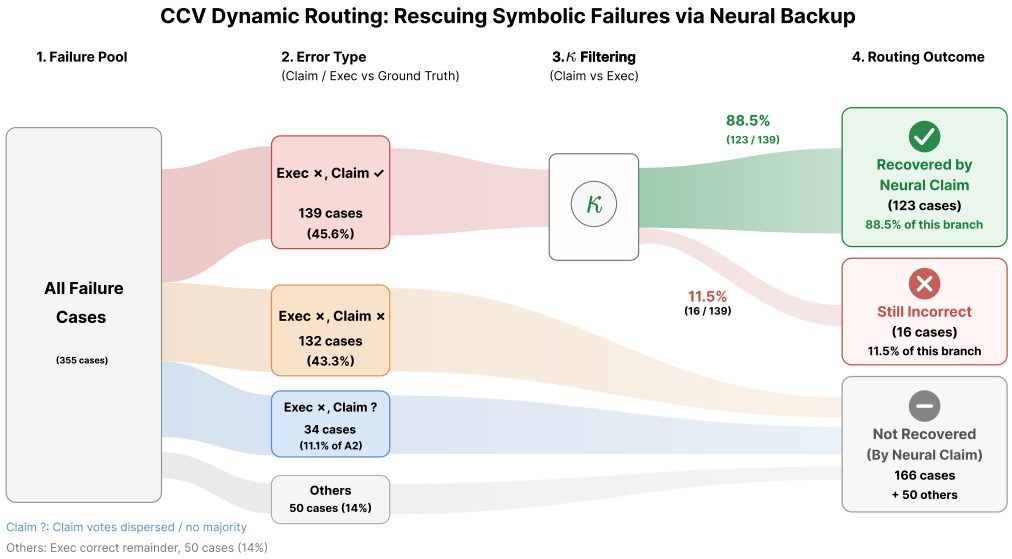


Figure D.1: **Error flow and dynamic routing.** Among 355 EAV failures with an oracle-correct candidate, wrong execution consensus dominates (86%) and splits into three subtypes: Exec✗ Claim✓, Exec✗ Claim✗, and Exec✗ Claim?. In the recoverable Exec✗ Claim✓ stream, the claim majority is correct while the execution majority is wrong; CCV's κ filter routes 123/139 cases (88.5%) to the correct neural-claim output, illustrating how output-pathway routing rescues samples when the symbolic channel collapses.

Table E.1: Cross-model validation on the same frozen $K{=}16$ four-benchmark protocol ($N{=}1709$), using $\tau_{\text{exec}}{=}9$ for both models. Absolute accuracy differs because Qwen3.5-9B has higher model capacity than Gemma4-E4B, but the gain pattern is consistent: cross-layer κ filtering and CCV improve over surface consensus for both models without additional inference.

Method	Qwen3.5-9B	Gemma4-E4B
Claim MV	70.57	44.94
Exec MV	74.31	41.08
EAV	76.18	44.88
PROVE-style κ filter	79.52	51.90
CCV (ours)	**80.98**	**52.78**
CCV − Claim MV	+10.41	+7.84
CCV − EAV	+4.80	+7.90
Oracle	93.27	91.75

I Qualitative Case Studies

We present four representative cases from the frozen Qwen3.5-9B s240 $K{=}16$, $T{=}0.8$ pool, selected to illustrate the four behavioral modes of CCV: (1) successful rescue when execution consensus is wrong but claim is correct, (2) hard cases where both pathways agree on a wrong answer, (3) correlated perception errors that produce incorrect agreement, and (4) fallback routing when no self-consistent candidate exists. Each case reports method outputs and the key DSL of the CCV-selected candidate.

I.1 CCV Rescue (Exec✗ Claim✓)

`geometry3k_test` qid 325. Expected: B (16). Choices: A=12, B=16, C=51, D=65.

Method outputs. Claim MV: B; Exec MV: C; EAV: C; CCV: **B**.

CCV-selected DSL ($\kappa = 1$, claim cluster 9, exec cluster 6):

Table F.1: Gemma4-E4B K scaling from the frozen K=16 pool. Entries for K=4, 8 are mean±std over four subsamples.

K	Claim MV	Exec MV	EAV	PROVE-style κ	CCV	Oracle	CCV − Claim MV
4	42.41±0.81	37.60±0.42	46.94±0.46	46.65±0.74	47.28±0.60	72.24±0.60	+4.87
8	44.00±0.48	39.86±0.24	48.82±0.79	50.04±0.62	50.85±0.58	83.84±0.32	+6.85
16	44.94	41.08	44.88	51.90	52.78	91.75	+7.84

Table H.1: Sensitivity to the candidate-level relative tolerance in $X_i \approx Y_i$ on the frozen Qwen3.5-9B s240 K=16, T=0.8 pool.

Setting	Candidate relative tolerance	CCV accuracy	κ candidates / precision
10^{-6}	0.000001	80.98	15,535 / 87.07
10^{-4}	0.0001	80.98	15,535 / 87.07
10^{-3}	0.001	80.98	15,535 / 87.07
default (10^{-2})	0.01	80.98	15,535 / 87.07
5×10^{-2}	0.05	80.98	15,535 / 87.07

```
angle_X = angle_L
x = angle_X / 4
answer = x          # 16.25 -> B
```

CCV selects a κ-consistent candidate whose claim and execution agree on B, while EAV follows a wrong execution majority (C). This is the canonical recoverable failure: the neural pathway is correct but the symbolic pathway is corrupted by shared perception error.

I.2 Hard Case: Exec✗ Claim✗

`geometry3k_test` qid 72. Expected: A (3). Choices: A=3, B=4, C=6, D=7.

Method outputs. Claim MV: B; Exec MV: B; EAV: B; CCV: B.

CCV-selected DSL ($\kappa = 1$, claim cluster 15, exec cluster 15):

```
x = 4
answer = x          # 4.0 -> B
```

All methods agree on the wrong answer. This case sits outside selection-only verification because the cross-channel agreement is itself wrong—both pathways share the same perceptual error.

I.3 Correlated Perception Error

`geometry3k_test` qid 35. Expected: D (155). Choices: A=20, B=25, C=30, D=155.

Method outputs. Claim MV: B; Exec MV: B; EAV: B; CCV: B.

CCV-selected DSL ($\kappa = 1$, claim cluster 16, exec cluster 16):

```
arc_BC = 30
arc_AD = 20
angle_2 = (arc_BC + arc_AD) / 2
answer = angle_2    # 25.0 -> B
```

This is a boundary case for CCV: a correlated visual error makes both channels self-consistent around the wrong answer. With 16/16 candidates in both clusters, selection alone has no alternative evidence to break the incorrect consensus.

I.4 Empty κ-Pool with Execution Fallback

`geometry3k_test` qid 56. Expected: C (70). Choices: A=66, B=68, C=70, D=72.

Method outputs. Claim MV: D; Exec MV: C; EAV: C; CCV: **C**.

CCV-selected DSL ($\kappa = 0$, claim cluster 10, exec cluster 16):

```
leg1 = 24
hypotenuse = 74
x = sqrt(hypotenuse**2 - leg1**2)
answer = x          # 70.0 -> C
```

The κ-pool is empty (no candidate's claim matches its execution), so CCV falls back to the execution-majority pathway. The large exec cluster (16/16) exceeds τ_{exec}, providing the correct answer. This illustrates dual-pathway routing: when cross-channel agreement is absent, the deterministic symbolic path remains available as a fallback.

J Proofs for the Cross-Channel Verification Analysis

J.1 Latent shared-perception model

We formalize the verification problem using answer-valued random variables. Let X_i denote the claim answer and Y_i the executed answer for candidate i, both taking values in an answer space $\mathcal{Z}$ with correct answer z^*. A latent visual state V is shared across candidates. Conditional on V, the executed answers Y_i are i.i.d. from $P_e(\cdot \mid V)$; the claim answers X_i are i.i.d. from $P_c(\cdot)$, independent of V; and X_i and Y_i are conditionally independent given V. This is an idealized model: empirically, we only require false self-consistency to be sufficiently rare, not perfect independence.

Define $p_e = P(Y_i = z^*) = \mathbb{E}_V[P_e(z^* \mid V)]$ and $p_c = P(X_i = z^*) = P_c(z^*)$. For execution correctness indicators $E_i = \mathbf{1}[Y_i = z^*]$, the within-channel execution correlation is

$$\rho_e = \mathrm{Corr}(E_i, E_j) = \frac{\mathrm{Var}_V[P_e(z^* \mid V)]}{p_e(1 - p_e)}. \tag{J.1}$$

By the law of total covariance applied to $E_i = \mathbf{1}[Y_i = z^*]$ for $i \neq j$, $\mathrm{Cov}(E_i, E_j) = \mathrm{Var}_V[P_e(z^* \mid V)]$, so shared perception creates positive correlation whenever execution accuracy varies with the visual state.

J.2 Proof of Theorem 1

Let $\kappa_i = \mathbf{1}[X_i = Y_i]$. Decompose self-consistency into true and false agreement:

$$t_\kappa = P(X_i = Y_i = z^*), \tag{J.2}$$

$$f_\kappa = P(X_i = Y_i \neq z^*) = \sum_{z \neq z^*} P(X_i = z, Y_i = z). \tag{J.3}$$

Then $P(\kappa_i = 1) = t_\kappa + f_\kappa$, and by Bayes rule:

$$P(Y_i = z^* \mid \kappa_i = 1) = \frac{t_\kappa}{t_\kappa + f_\kappa}. \tag{J.4}$$

This identity does not require independence; it simply states that self-consistent candidates are correct when agreement is true rather than false.

Under the latent shared-perception model, conditional independence of X_i and Y_i given V yields $t_\kappa = P_c(z^*)\,\mathbb{E}_V[P_e(z^* \mid V)] = p_c p_e$. If wrong answers are approximately dispersed uniformly over $M-1$ alternatives, then

$$f_\kappa \approx \frac{(1 - p_c)(1 - p_e)}{M - 1}. \tag{J.5}$$

The condition $p_\kappa > p_e$ reduces to $p_c(1 - p_e) > f_\kappa$. Substituting the uniform-dispersion approximation gives $p_c > (1 - p_c)/(M - 1)$, i.e., $p_c > 1/M$ for $p_e < 1$. Thus cross-channel agreement improves per-candidate precision whenever the claim pathway is better than random over the effective answer space.

J.3 Proof of Theorem 2

For n exchangeable binary correctness variables with $P(B_i = 1) = p$ and pairwise correlation ρ, the variance of the sample mean is $p(1-p)[1 + (n-1)\rho]/n$. Applying the SNR definition from the main text to raw exec voting (p_e, K, ρ_e) and to the retained κ-filtered pool $(p_\kappa, K_\kappa, \rho_\kappa)$ gives $\mathrm{SNR}_\kappa > \mathrm{SNR}_{\mathrm{raw}}$ iff

$$\frac{p_\kappa - \frac{1}{2}}{p_e - \frac{1}{2}} > \sqrt{\frac{p_\kappa(1-p_\kappa)}{p_e(1-p_e)} \cdot \frac{K(1+(K_\kappa-1)\rho_\kappa)}{K_\kappa(1+(K-1)\rho_e)}}. \tag{J.6}$$

Corollary 1 (Answer dispersion amplifies cross-channel advantage)**.** *Under the uniform-dispersion approximation, as the effective answer space M grows, $f_\kappa \to 0$ and hence $p_\kappa \to 1$. If the retained pool remains nonempty and the correlation factor remains bounded, then $\mathrm{SNR}_\kappa \to \infty$ while raw exec SNR is unchanged.*

J.4 Numerical illustration

We instantiate the SNR proxy using measured post-hoc statistics from the full dataset. Failed executions are counted as incorrect; correlations are Pearson correlations over ordered within-question candidate pairs. Raw exec voting uses $K = 16$, $p_e = 0.617$, and $\rho_e = 0.526$. The retained pool has mean size $K_\kappa = 9.09$, $p_\kappa = 0.871$, and $\rho_\kappa = 0.646$. Table J.1 summarizes the resulting SNR proxy values.

Table J.1: Measured statistics for the one-vs-rest SNR proxy.

Method	n	p	ρ	SNR
Raw exec MV	16	0.617	0.526	0.32
κ-filtered MV	9.09	0.871	0.646	1.33

The retained pool is therefore 23% more correlated, but the SNR proxy improves by $4.1\times$. This empirical regime is exactly the counterintuitive case captured by Theorem 2: precision filtering dominates the correlation penalty. The observed false self-consistency rate is 12.9%, so most κ=1 candidates correspond to true rather than accidental agreement.

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